

The optimization problem is solved by the limited Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) method which requires the objective function and the gradients of the objective function with respect to the optimization variables. The gradients of the objective function with respect to the optimization variables are obtained by the adjoint method which has been thoroughly discussed in [27]. The flow chart of the procedure to solve the inverse problem is shown in Fig. 5.

To test the feasibility of the proposed inverse scheme, we adopt the displacement datasets generated by a forward finite element simulation with a target shear modulus distribution. The white Gaussian random is also added into the simulated surface displacement field. The noise

level is defined by $\sqrt{\sum_{k=1}^{N_{\text{node}}} (u_k - u_k^{\text{noise}})^2} / \sqrt{\sum_{k=1}^{N_{\text{node}}} (u_k)^2} \times 100\%$, where u_k and u_k^{noise} are the simulated and noisy displacement of k -th node, N_{node} is the total number of nodes in interest domain. In comparison, we also solve inverse problems using the prevalent implicit inverse method based on the adjoint method where the shear moduli are nodally defined [22]. To quantitatively analyze the performance of the proposed method, we define the average relative error by $\frac{1}{N_{\text{node}}} \sum_{k=1}^{N_{\text{node}}} \left| \left(\mu_k - \mu_k^{\text{target}} \right) / \mu_k^{\text{target}} \right| \times 100\%$, where μ_k is the recovered shear modulus value of k -th node and μ_k^{target} is the target shear modulus value of k -th node.

Gradient computation

To efficiently compute the gradient of the objective function with respect to the optimization variable, we have adopted the adjoint method, which involves solving the associated adjoint equation. The adjoint equation is given by [34,35]:

$$\mathcal{B}(\mathbf{w}^h, q^h, \delta \mathbf{u}^h, \delta p^h; \mathbf{u}^h, p^h) = -(\delta \mathbf{u}^h, \mathbf{u}^h - \mathbf{u}^m)_{\partial \Omega}, \quad \forall [\delta \mathbf{u}^h, \delta p^h] \in \mathcal{M}^h \times \mathcal{P}^h \quad (16)$$

where $\mathcal{B}(\mathbf{w}^h, q^h, \delta \mathbf{u}^h, \delta p^h; \mathbf{u}^h, p^h) = \frac{d}{d\varepsilon} \mathcal{A}^*(\mathbf{w}^h, q^h, \mathbf{u}^h + \varepsilon \delta \mathbf{u}^h, p^h + \varepsilon \delta p^h) \big|_{\varepsilon=0}$ is the linearization of $\mathcal{A}^*$. With the acquired state variables $\mathbf{u}^h, p^h$, we can solve for the dual variables $\mathbf{w}^h, q^h$ using Eq. (16). To this end, the gradient of the objective function with respect to the shear modulus:

$$\delta \Pi = D_{\mu} \Pi \delta \mu = \mathcal{C} + \alpha \int_{\Omega} \frac{\nabla \mu \cdot \nabla \delta \mu}{\sqrt{|\nabla \mu|^2 + c^2}} d\Omega \quad (17)$$

$\mathcal{C} = \frac{d}{d\varepsilon} \mathcal{A}^*(\mathbf{w}^h, q^h, \mathbf{u}^h, p^h, \mu + \varepsilon \delta \mu) \big|_{\varepsilon=0}$. This can be used to efficiently compute the gradient of the objective function with respect to

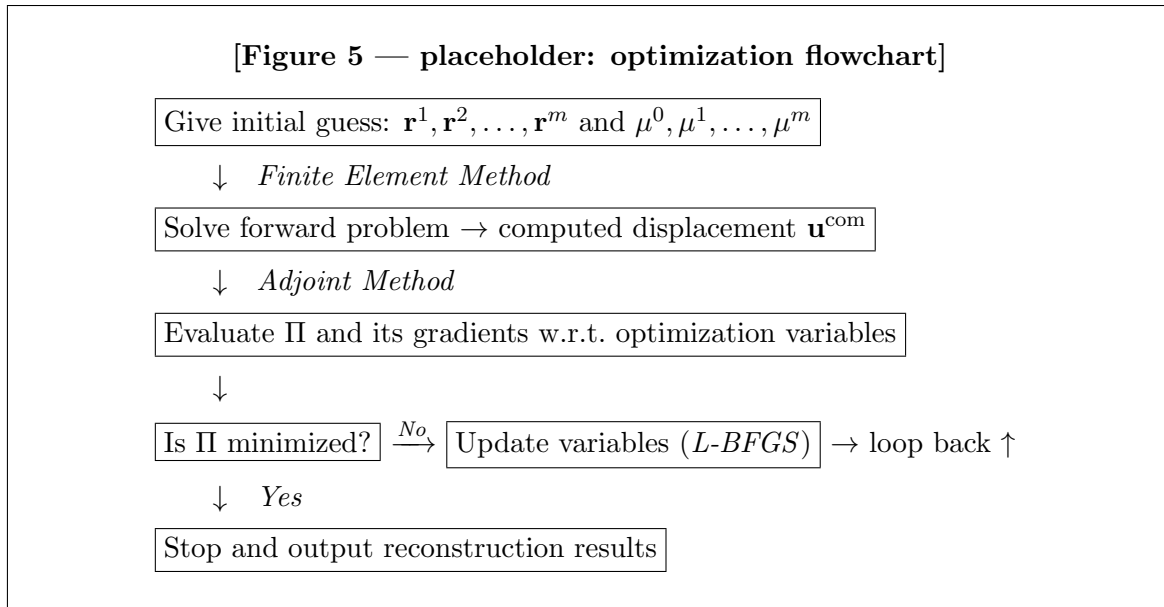


Fig. 5. The flow chart of the proposed explicit inverse method.